



Hellen Yu
FLOWCOM MECHANICAL TECHNOLOGY (SHANGHAI) CO LTD
UNIT 2-801, NO. 2388 XIUPU ROAD,
PUDONG
SHANGHAI
201315 CHINA

Date: 2019/06/03
Subscriber: 100541872
PartySite:
File No:
Order No: 12820142
Project Number: 19OR12820142
PD No: 19029406
PO Number:

Subject: Multiple Listing Correlation Sheet

Dear Hellen Yu,

Enclosed is a copy of your completed Multiple Listing Correlation Sheet covering the Basic File listed below. The following documents were added (new) or revised on 06/04/2019.

<u>Basic File</u>	<u>Vol</u>	<u>ML File</u>	<u>Pages</u>	<u>Revised</u>	<u>New</u>
	1	EX28079	New ML Correlation Page(s)		

Please review this material and report any inaccuracies to YA-PING YU (EXT. 6702) referring to the above Order number.

This material is provided on behalf of UL LLC(UL) or any authorized licensee of UL.

SUZ File



Multiple Listing Correlation Sheet

ML FILE NO. EX28079

Issued: 2019-06-04

MULTIPLE LISTING
of
SPRINKLERS, AUTOMATIC AND OPEN
(VNIV)
for

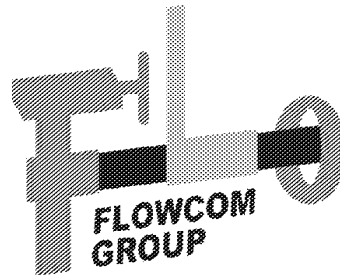
[610631] FLOWCOM MECHANICAL TECHNOLOGY
(SHANGHAI) CO LTD

Basically Listed products covered by Procedure and/or Reports

Products Covered	Report Date		Multiple Listee's Product Designation	ILL #
STANDARD SIDEWALL- SPRAY SPRINKLERS	2018-12-20		FW1205	1
			FW2205	1
UPRIGHT AND PENDENT STANDARD-SPRAY SPRINKLERS	2018-12-20		FW1005	1
			FW2005	1
			FW1105	1
			FW2105	1
CONVENTIONAL (OLD STYLE) SPRINKLERS	2018-12-20		FW1305	1

MARKING: Same as that described in Follow-Up Service Procedure and/or Report except for Multiple Listee's name, file number, Trademark / Tradename, **FLO** when applicable, and product designation. The Control Number remains [5MJ6].

LITERATURE: If literature is packaged with the Multiple Listed product(s) it shall be in compliance with the requirements outlined in the appropriate UL Standard(s).



FlowCom Mechanical Technology (Shanghai) Co., Ltd.

SPRINKLER TECHNICAL DATA

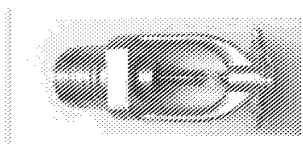
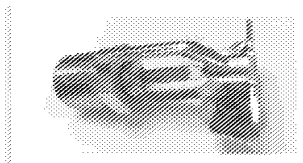
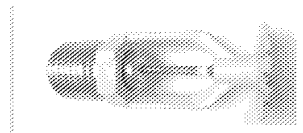
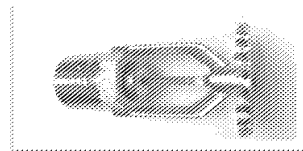
N191935218

DESCRIPTION

Pendent and upright and conventional and sidewall sprinklers are designed for use in automatic sprinkler systems designed in accordance with standard installation (NFPA13). The temperature response is standard response 5mm glass bulb and fast response 3mm glass bulb. The sprinklers come with compact design, available in different finishes and temperature ratings.

The sprinklers body is made of brass die-casting copper alloy (DZR). The spring seat consists of a spring Belleville washer coated on both side with a Teflon tape.

Pendent and conventional and sidewall sprinklers are installed with adjustable or non-adjustable escutcheon plates. These are intended to be installed finished ceilings. Escutcheon plate adjustment provides convenience in pendent and conventional sprinkler installation in case of areas with finished ceiling or walls.



TECHNICAL SPECIFICATION

Style	Pendent, Upright, Sidewall, Conventional and Recessed Pendent, Recessed Sidewall, Recessed Conventional
K Factor gpm/(psi) ^{1/2} (L/min(bar) ^{1/2})	5.6 (80)
Nominal Thread Size	1/2" NPT OR 1/2" BSPT
Max. Working Pressure	175 PSI (12BAR)
Factory Test Pressure	500 PSI (35BAR)
Min. Operating Pressure	7 PSI (0.5 BAR)
Finish	Brass and Chrome Plated and White Plated Available

TECHNICAL SPECIFICATIONS

SIN	Response type	Sprinkler Style	Temp. Rating F (C)
-----	---------------	-----------------	--------------------

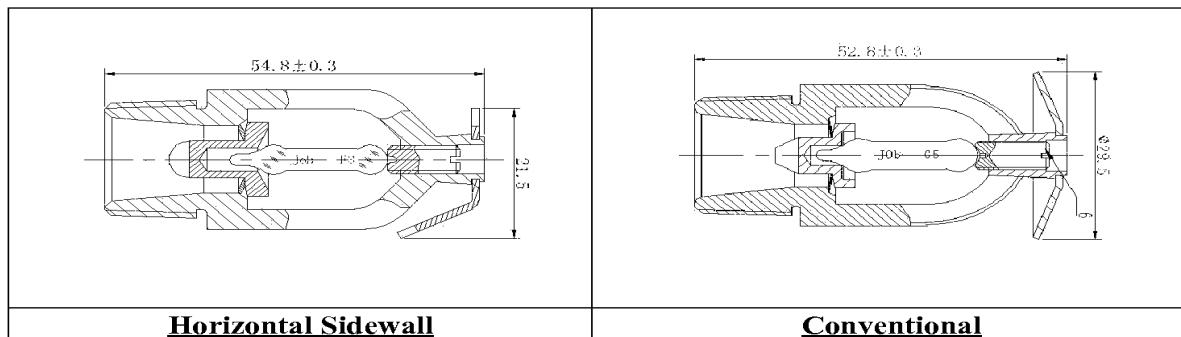
FW1205	SR	Sidewall	135(57) 155(68) 175(79)
FW2205	QR	Sidewall	155(68)
FW1005	SR	Upright	135(57) 155(68) 175(79)
FW2005	QR	Upright	155(68)
FW1105	SR	Pendent	135(57) 155(68) 175(79)
FW2105	QR	Pendent	155(68)
FW1305	SR	Conventional	155(68)

Material Specification

Deflector	Brass UNS-28000
Bulb	Glass with Glycerin Solution JOB G5 or Day-impex-937 for FW1205, FW1005, FW1105, FW1305 JOB F3 or Day-impex-941 for FW2205, FW2005, FW2105
Load Screw	Brass UNS-28000
Seal	Brass UNS-28000
Frame	Brass UNS-28000
Seal Washer	Belleville Washer coated on both sides with Teflon Tape
Bulb Nominal Diameter	Standard Response 5.0mm, Quick Response 3.0mm

Each style sprinkler dimension

mm	
<u>Pendent</u>	<u>Upright</u>

**INSTALLATION**

The sprinklers which are manufactured and tested in accordance with the rigid requirements of the Standard UL 199, also should be installed in accordance with the current Standard NFPA 13. The system piping must be properly sized to insure the minimum required flow rate at the sprinkler. Check for the proper model, style, orifice size and temperature rating prior to installation, install sprinkler after the piping is in place to avoid mechanical damage, replace any damage units, wet pipe systems must be protected from freezing.

Upon completion of the installation, the system must be tested per recognized standard. In case of a thread leak, remove the unit, apply new pipe joint compound or tape and re-install.

INSTALLATION METHODS

- a. Pendent sprinkler must be mounted in a pendent position, and upright sprinkler must be mounted in upright position, the C U/P sprinkler may be mounted both in upright and pendent position, the horizontal sidewall sprinkler is install at horizontal position, to suit for field condition.
- b. Hand tighten the sprinkler into the sprinkler fitting. It is recommended that a torque of 7 ~14 ft-lbs be used to obtain a thread 1/2 inch NPT Sprinkler joint. A radial force of 10 – 20 lbs.
- c. For using only a non-hardening pipe joint compound or Teflon tape apply to the male thread only.
- d. Hand tighten the sprinkler into fitting, use a 22mm open spanner or 8" (200mm) adjustable wrench on flat to tighten the unit into the fitting, do not use spanner on the frame arms, for it will cause the arms break and glass bulb burst.

REMOVE PROTECTIVE COVER

Be sure to remove the plastic protective cover after completed the installation, should not clamp on the frame arms, otherwise will to prevent the heat response function.

➤ **Caution :**

A maximum of 21 ft-lbs of torque is to be used to install the sprinkler. Higher levels of torque may distort the sprinkler orifice seat saddle with consequent leakage.

➤ **Escutcheon for clean attractive installation**

In case of install a decoration of escutcheon for clean attractive purpose, the step as below:

1. Spin the inner piece of escutcheon into sprinkler threads.
2. Use only a non-hardening pipe joint compound or Teflon tape apply to the male thread only.
3. Hand tighten the sprinkler into fitting, use a 22mm open spanner or 8" (200mm) adjustable wrench to tighten the unit into the fitting.
4. Keep the plastic protective cover clamped on sprinkler arms while inner decoration work.
5. Once decoration work completed, take off the plastic protective cover.
6. Insert the outer piece of escutcheon into inner piece, make the escutcheon flush to the ceiling, then the sprinkler is ready for services.